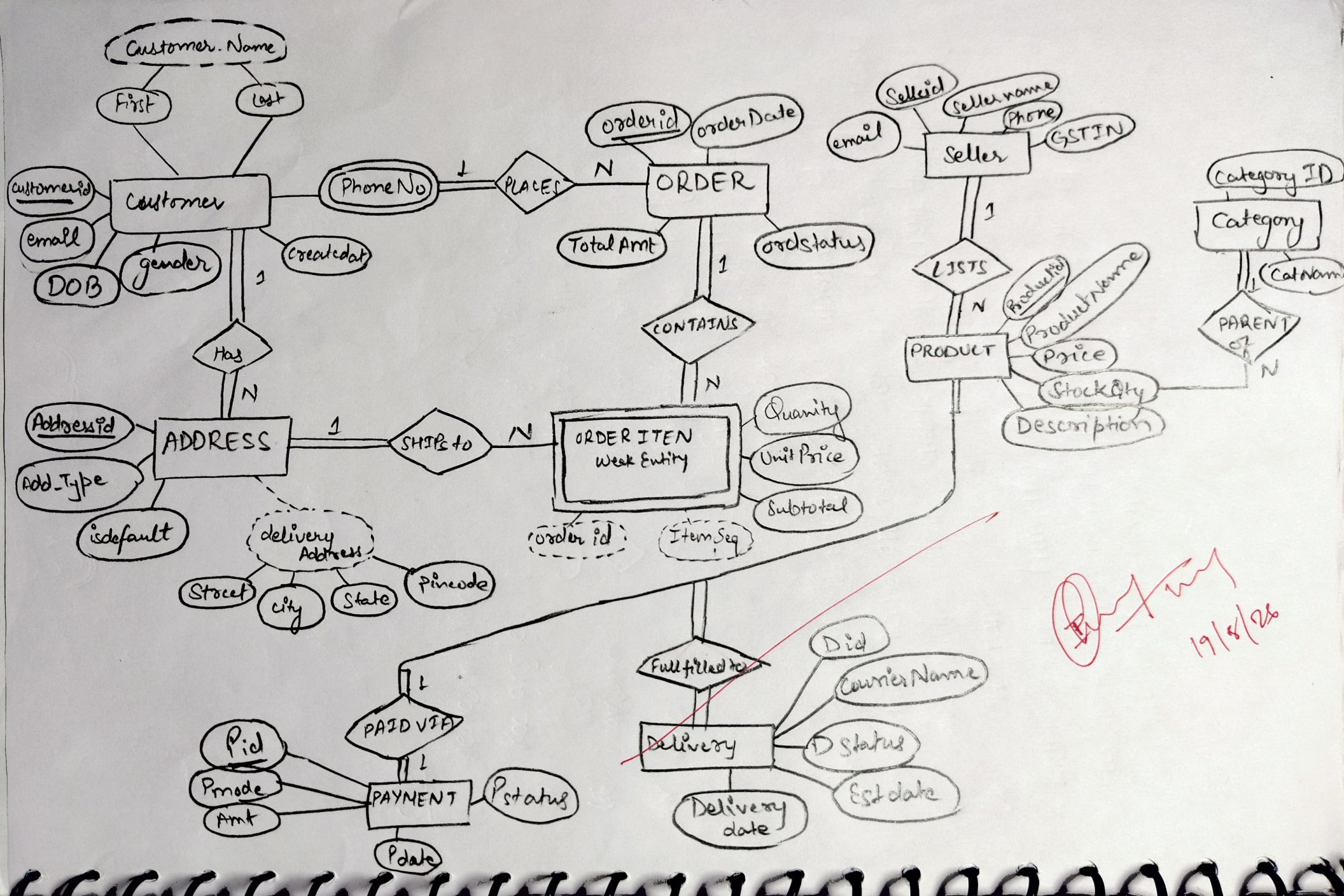

Customer. Name
First
Last
customerid
email
DOB
gender
Customer
createdat
PhoneNo
1
PLACES
N
ORDER
orderid
orderDate
Total Amt
ordstatus
1
CONTAINS
N
1
Has
N
Addressid
Add_Type
isdefault
ADDRESS
1
SHIPS to
N
ORDER ITEN
Weak Entity
Quanity
Unit Price
Subtotal
order id
Item Seq
delivery Address
Street
city
State
Pincode
Sellerid
seller name
Phone
email
GSTIN
Seller
1
LISTS
N
PRODUCT
Productid
Product Name
Price
StockQty
Description
Category ID
Category
1
Cat Nam
PARENT of
N
1
PAID VIA
1
Pid
Pmode
Amt
PAYMENT
Pstatus
Pdate
Fullfilled to
Delivery
Did
Courier Name
D status
Est date
Delivery date
19/8/26

Experiment: 1

Design an ER diagram for an Indian E-commerce platform with entities Customer, Product, Order, Order Item, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.

Code:-

Representation

(i) Rectangle — Entity in ER model

(ii) Ellipse — Attributes in ER model

(iii) Diamond — Relation among Entity

(iv) Line — Attributes to Entity and Entity Set with other relationship types.

An Entity-Relationship (ER) diagram is a conceptual representation of a diagram database. It shows entities, their attributes, relationships and the constraints between them.

1. Entities and Primary Keys

- Customer – CustomerID (PK), Email, DOB, Gender, CreatedAt, CustomerName.
- Address – AddressID (PK), AddressType, IsDefault and Delivery Address.
- Order – OrderID (PK), OrderDate, Orderstatus, Total Amount.

- OrderItem – Weak entity identified by OrderID + ItemSeq.
- Product – ProductID (PK), ProductName, Price, StockQty, Description.
- Seller – SellerID (PK), SellerName, Phone, Email, GSTIN.
- Category – CategoryID (PK), CategoryName.
- Payment – PaymentID (PK), PaymentMode, Amount, PaymentStatus, PaymentDate.
- Delivery – DeliveryID (PK), CourierName, TrackingNo, DeliveryStatus, EstimatedDate and DeliveredDate.

2. Relationships

- Customer – PLACES – Order: A customer can place many orders, while each other belongs to one customer (1:N).
- Customer – HAS – Address: A customer can have multiple addresses (1:N)
- Order – CONTAINS – OrderItem: An order contains one or more order items. OrderItem depends on Order.
- OrderItem – SHIPS TO – Address: An order item/ order is associated with the delivery address.
- Seller – LISTS – Product: A seller can list many products (1:N).
- Category – PARENT OF – Category: Categories can have sub-categories, forming a hierarchial relationship.
- Category – Product: Products are classified under categories.
- Order – PAIDVIA – Payment: An order is associated payment.
- Order – FULFILLEDBY – Delivery: An order is associated delivery.

3. Composite Attributes

A composite attribute can be divided into smaller meaningful attributes.

- CustomerName → FirstName + LastName
- DeliveryAddress → Street + City + State + Pincode

These are shown using a dashed oval connected to their component attributes.

4. Multi-Valued Attributes

A multi-valued attribute can have more than one value for the same entity.

- PhoneNo of Customer can have multiple phone numbers.
- TrackingNo is represented as multi-valued in the given daigram.

A multi-valued attribute is shown using a double oval.

5. Weak Entity

OrderItem is a weak entity because it cannot be uniquely identified without its related Order.

- Owner entity: Order
- Partial Key: ItemSeq
- Composite Primary Key: (OrderID, ItemSeq)

It is represented using a double rectangle and its identified through the CONTAINS relationship.

6. Participation Constraints

Participation specifies whether the existence of an entity in a relationship is mandatory or optional.

- Double line: Total participation / mandatory.
- Single line: Partial participation / optional.
- 1: One
- N: Many

For example, every Order must be associated with Customer, while a Customer may place zero or many Orders.